

## Project Architecture Requirements

### 1. Microservices

- The application must consist of at least **three independent backend services**.
- Each service must run in its own Docker container.
- Services must communicate **asynchronously** using **Apache Kafka**.
- Services may share a database or use separate databases, based on your design choice (monolithic DB vs decentralized DB is acceptable).

### 2. Example Use Cases (You can create your own)

- **E-commerce** (User Service, Order Service, Product Service)
  - **Hotel Booking** (Customer Service, Booking Service, Payment Service)
  - **Event Ticketing** (User Service, Ticket Service, Notification Service)
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## Technical Requirements

### 1. Backend

- You may use **any backend technology**, such as:
  - **Node.js (Express/NestJS)**
  - **ASP.NET Core**
  - **Python (FastAPI, Flask)**
  - **Java (Spring Boot)**
  - **Go, PHP, Ruby, etc.**
- Each service should expose its own **REST API**.

## 2. Kafka

- Use Kafka to publish and consume events between services.
- At least **two meaningful events** must flow between services (e.g., `OrderPlaced`, `UserRegistered`, `ProductOutOfStock`).

## 3. Docker

- All services must be containerized using Docker.
- Provide a `docker-compose.yml` to run the full system, including Kafka and any databases.

## 4. Frontend (Optional but Recommended)

- You may use any frontend technology:
  - **React, Angular, Vue.js**
  - Or even server-rendered pages via Razor (ASP.NET), EJS, etc.
- The frontend should interact with the backend services via **API Gateway** or **direct REST calls** to the services.